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(54) **METHOD FOR MANUFACTURING AN ALUMINUM NITRIDE OPTICAL WAVEGUIDE, AND ALUMINUM NITRIDE OPTICAL WAVEGUIDE**

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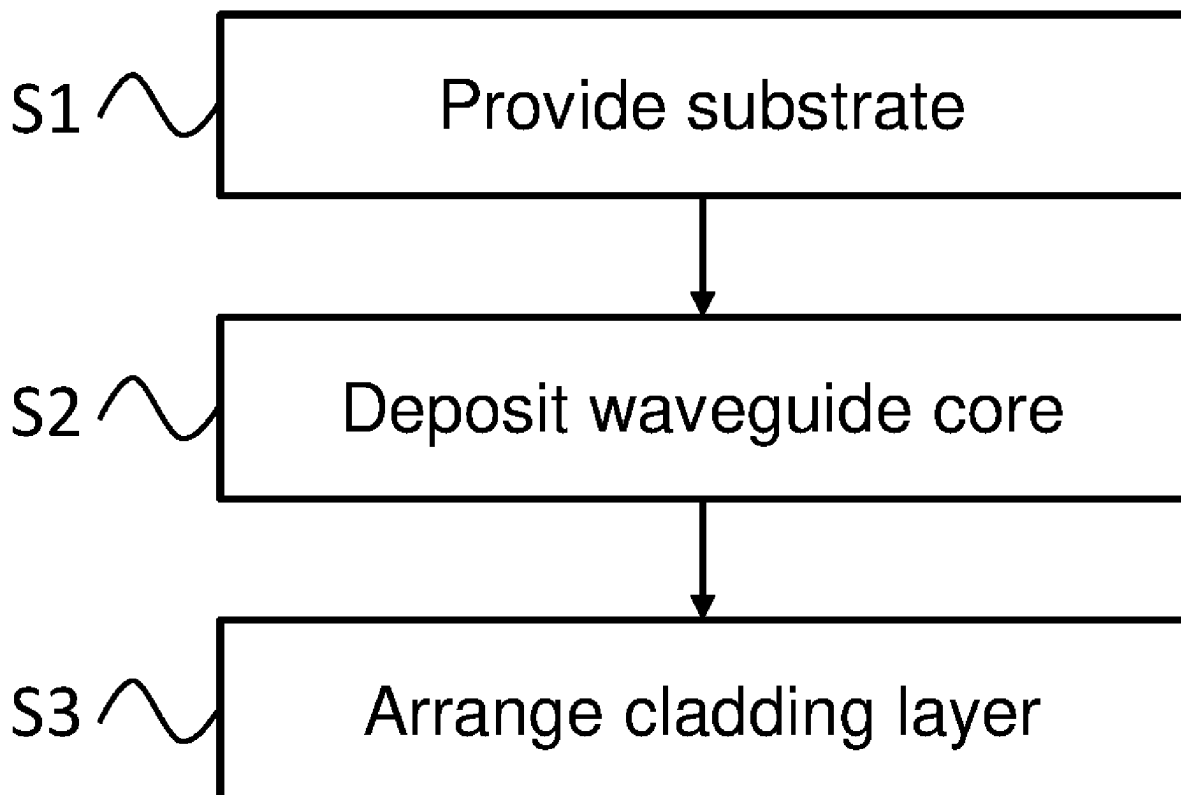
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(57) **ABSTRACT**

The present application concerns a method for manufacturing an optical waveguide. The method comprising providing a substrate, depositing an aluminium nitride waveguide core on the substrate, and arranging a cladding layer on the deposited aluminium nitride waveguide core. The arranging comprising at least one processing step during which the deposited aluminium nitride waveguide core is subjected to a given maximum temperature. The method is characterised in that depositing the aluminium nitride waveguide core comprises forming nano-crystallites in the aluminium nitride waveguide core, and in that the temperature the aluminium nitride waveguide core would be required to have for significantly increasing a size of the formed nano-crystallites during arranging of the cladding layer exceeds the given maximum temperature, wherein the given maximum temperature is about 800 degrees Celsius or higher.



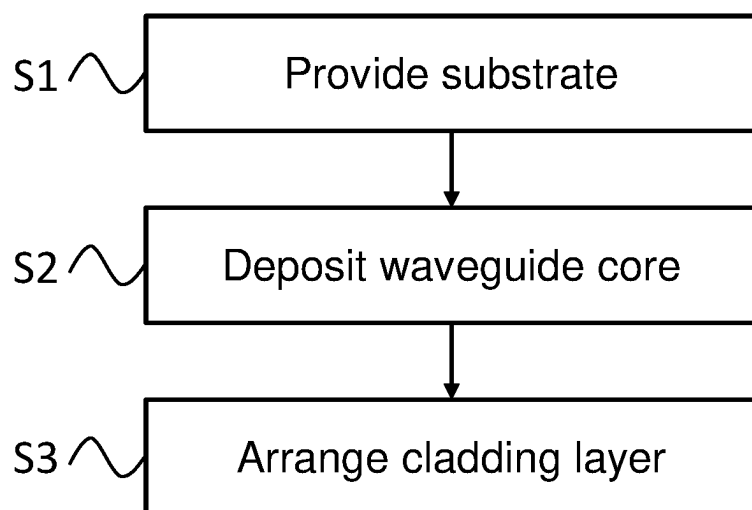


FIG. 1

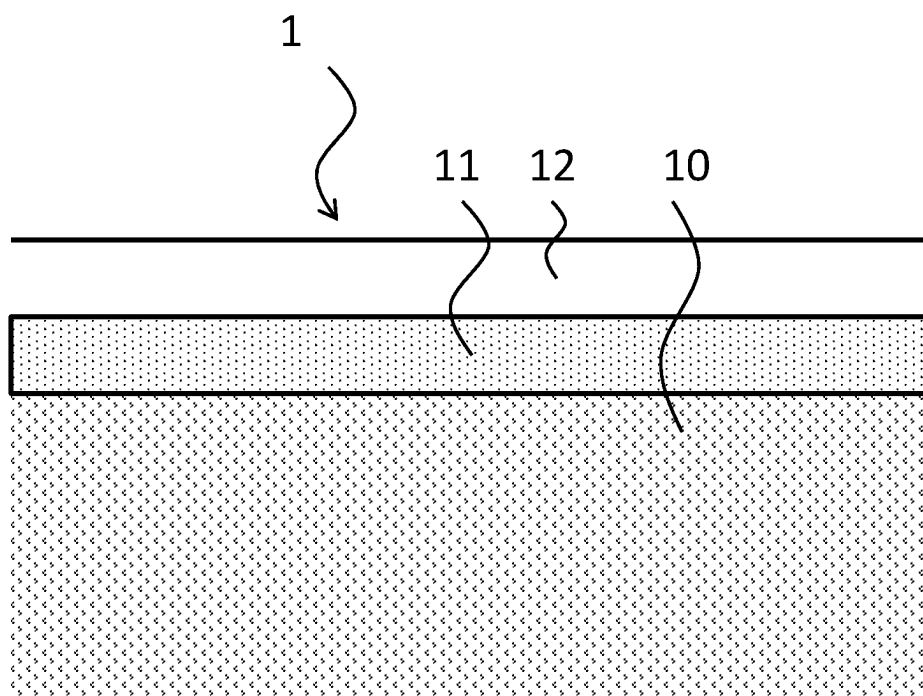


FIG. 2

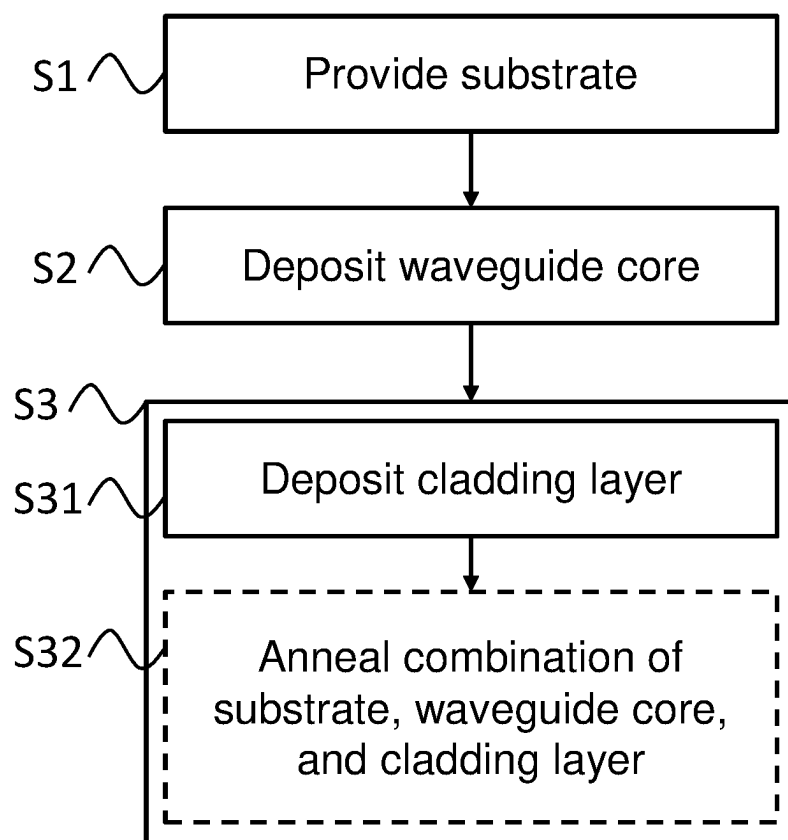


FIG. 3

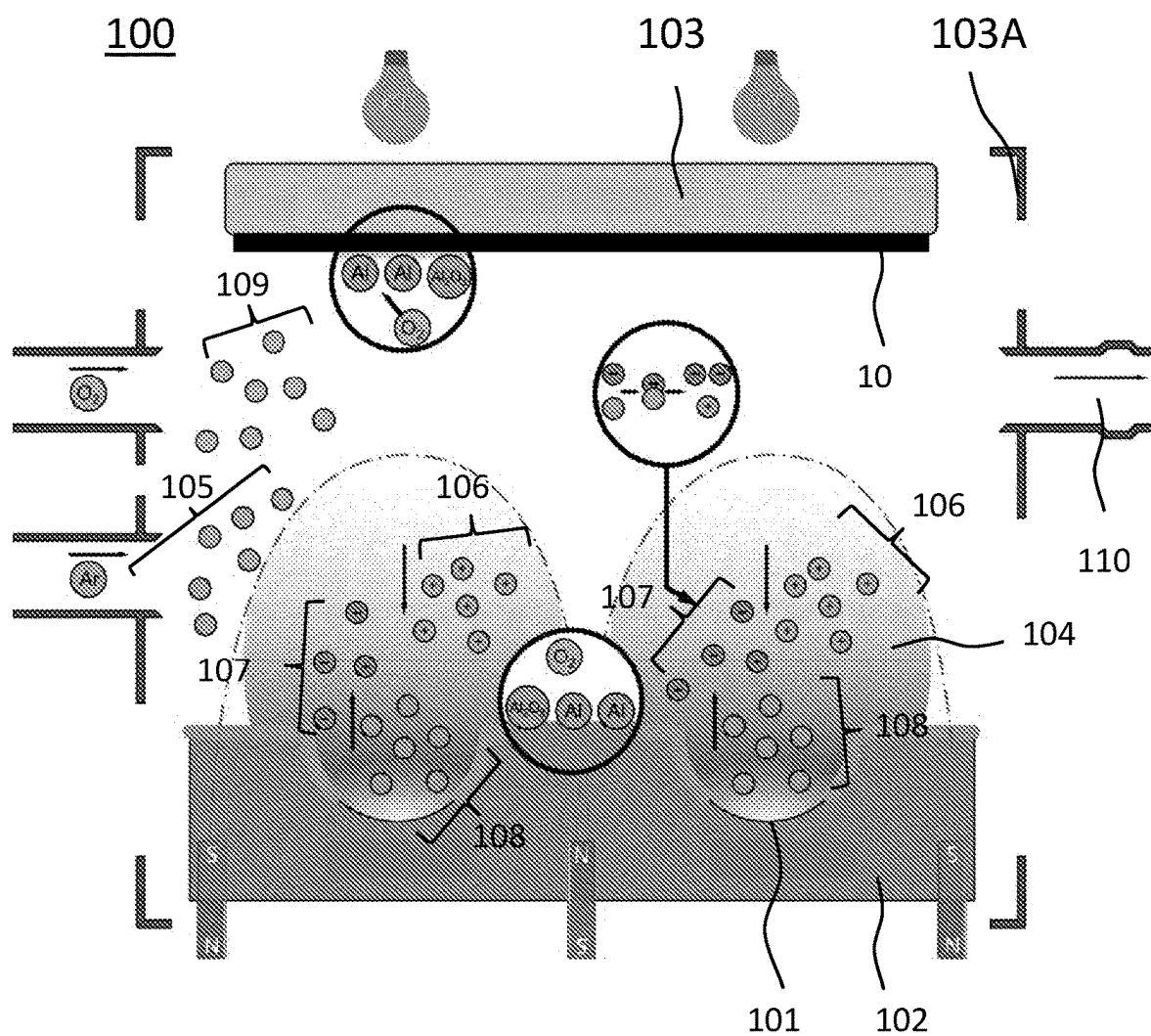


FIG. 4

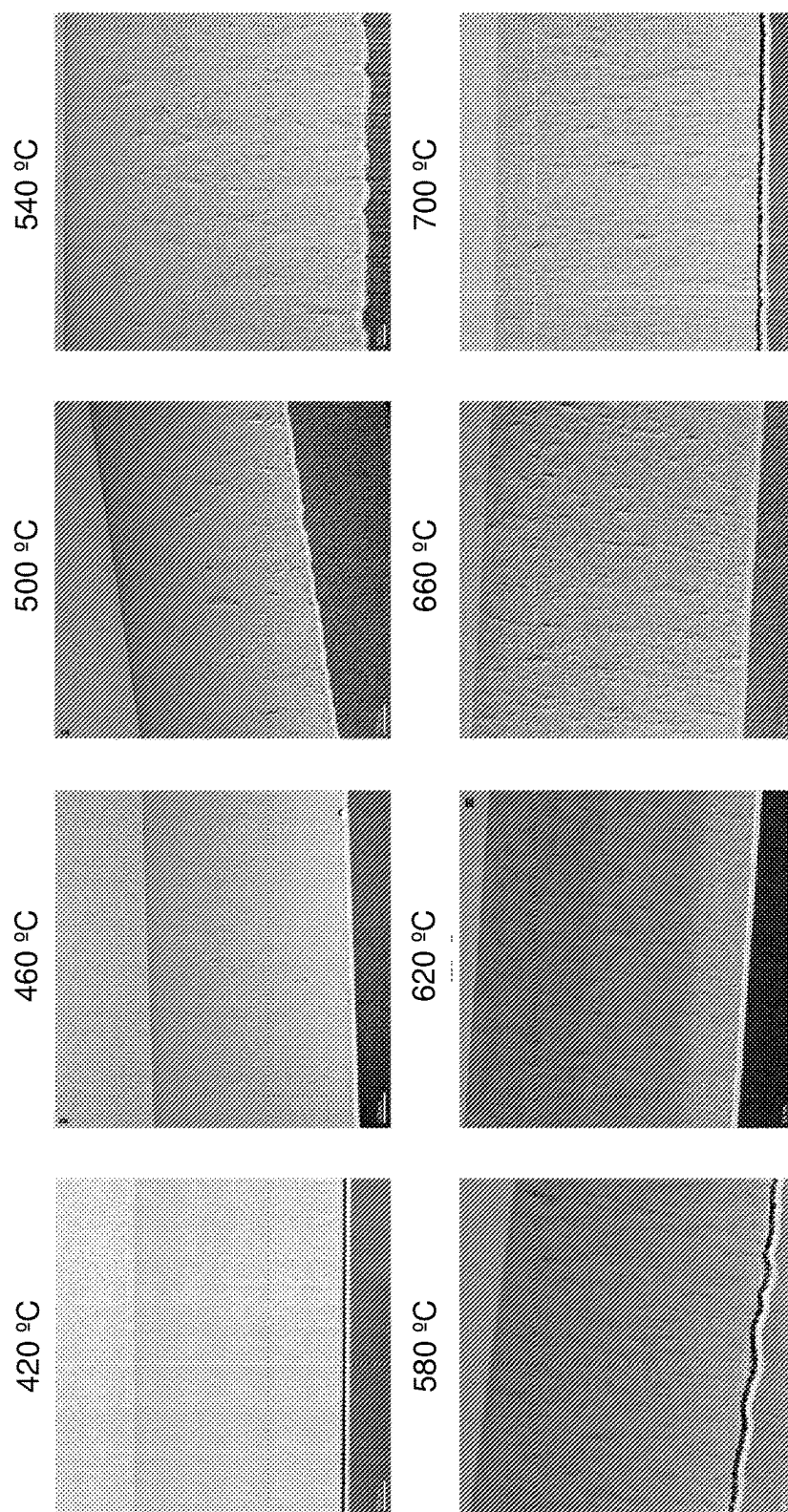


FIG. 5

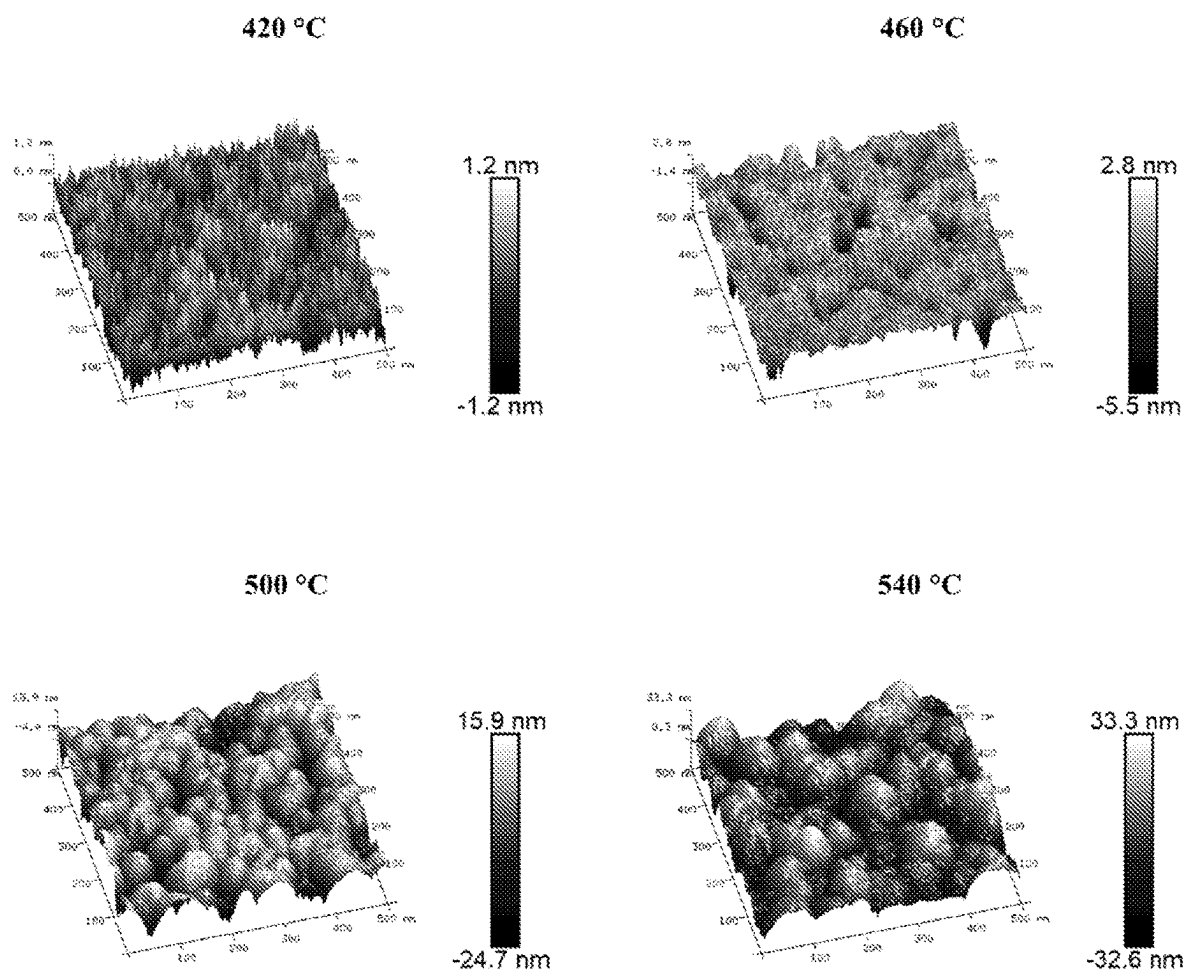


FIG. 6a

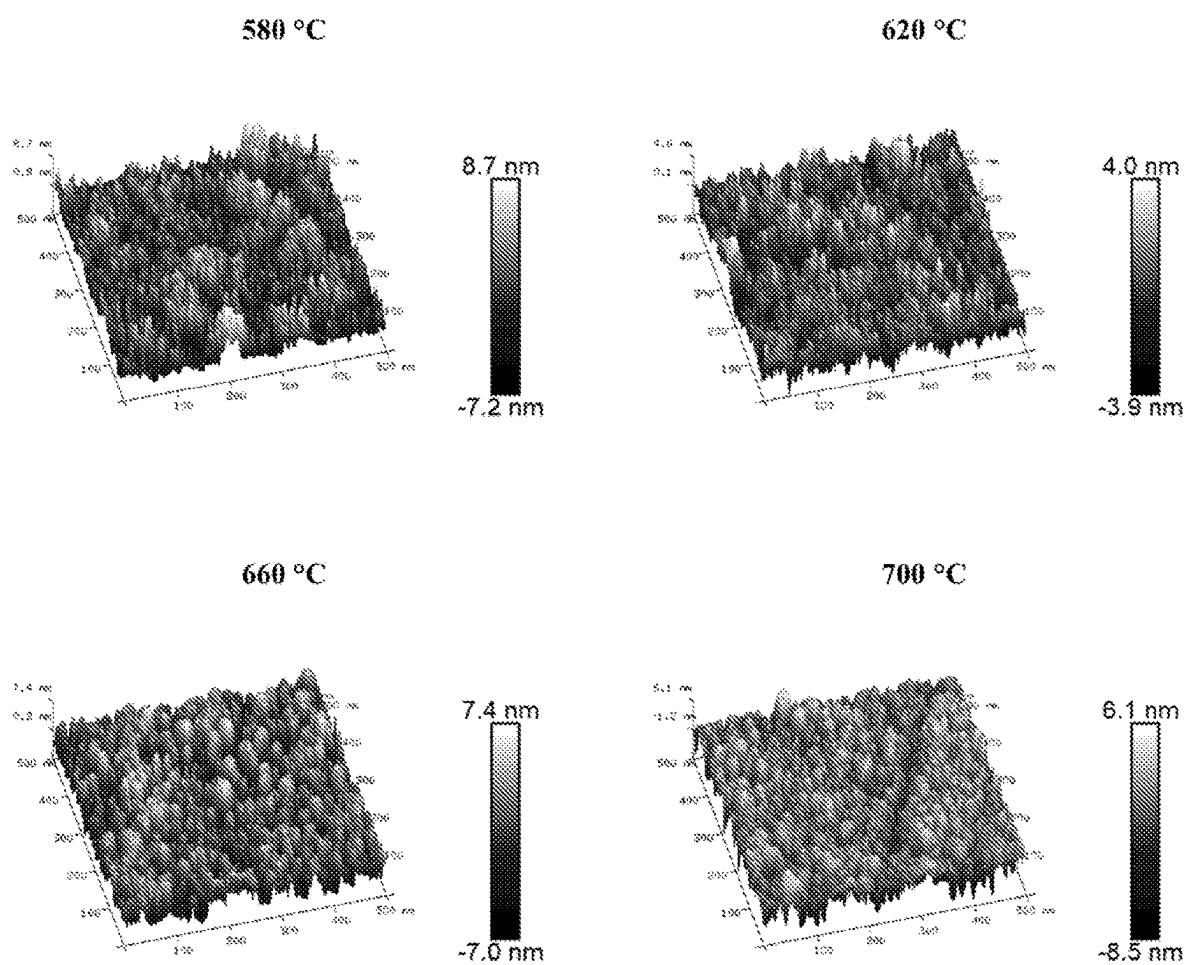


FIG. 6b

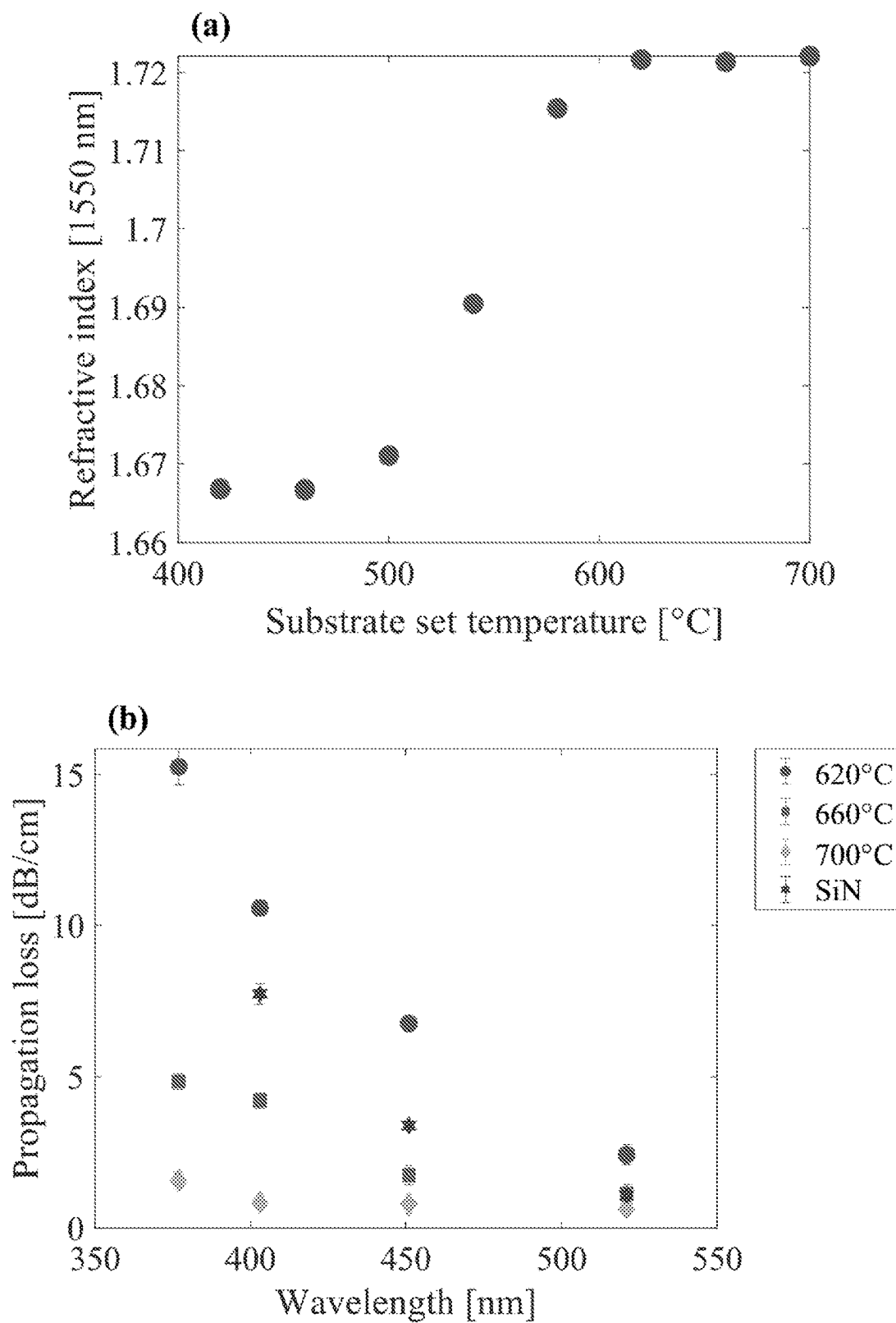


FIG. 7

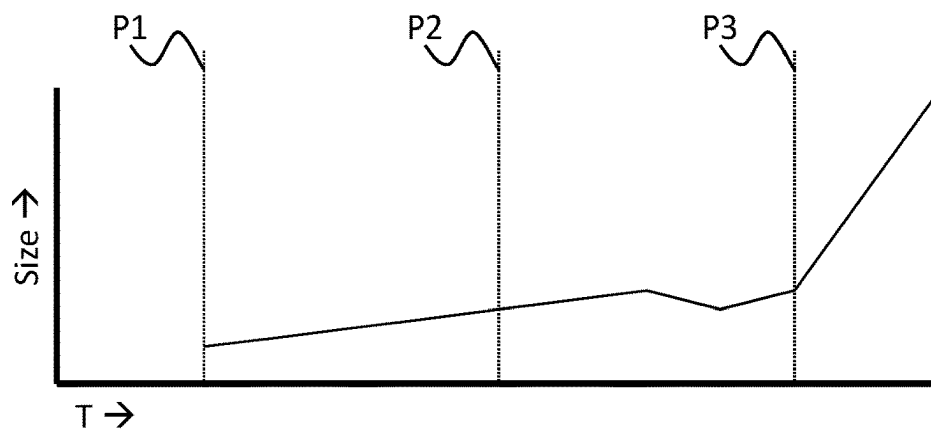


FIG. 8a

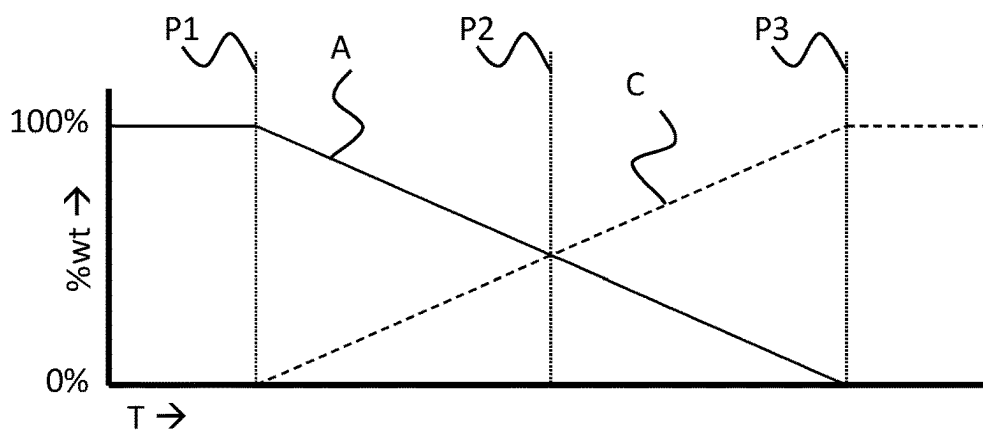


FIG. 8b

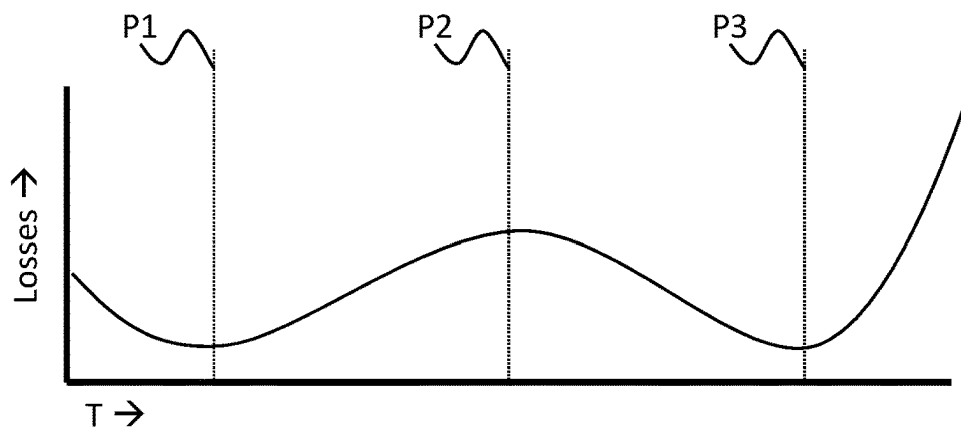


FIG. 8c

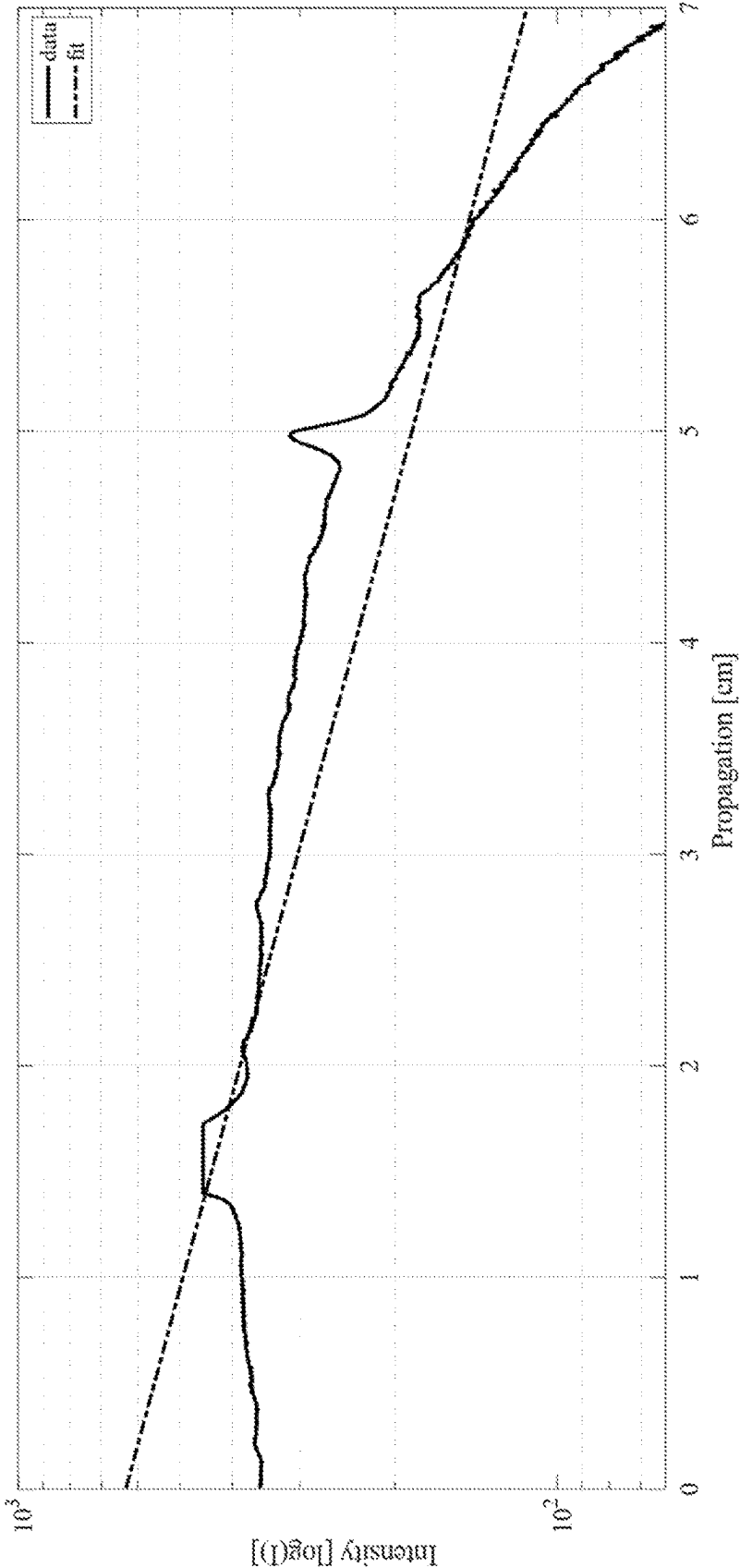


FIG. 9a

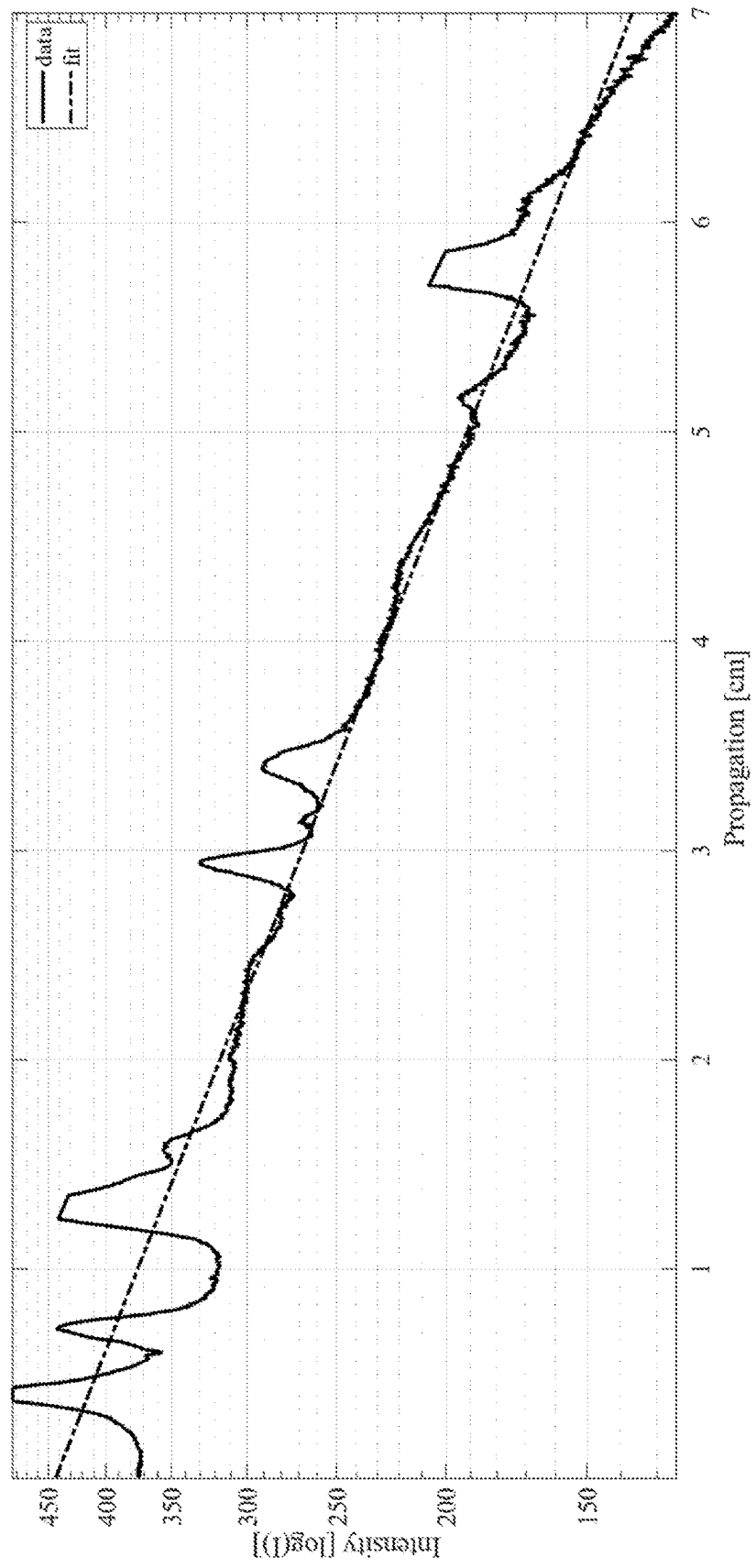


FIG. 9b

METHOD FOR MANUFACTURING AN ALUMINUM NITRIDE OPTICAL WAVEGUIDE, AND ALUMINUM NITRIDE OPTICAL WAVEGUIDE

[0001] The present application concerns a method for manufacturing an optical waveguide. The present application further concerns an optical waveguide, preferably manufactured by such method.

[0002] Integrated photonics has become ubiquitous as its development opens opportunities for improved capabilities in applications over well-developed microelectronic technologies. Especially, as ultra-low loss waveguides are realized, a myriad of applications for photonics integrated circuits can be explored, among others, quantum computing, microwave photonics, biosensing, and non-linear sources. Of the materials investigated for integrated photonics, aluminium oxide, Al_2O_3 , has emerged as a promising platform material due to its large transparency window, low propagation losses, and high rare-earth solubility.

[0003] Another material investigated is aluminium nitride, AlN , in part due to its transparency window covering more of the UV spectrum (e.g. down to 150 nanometre, compared to 200 nanometre for aluminium oxide).

[0004] Low losses of $0.04+0.02$ dB/cm in the C-band have been demonstrated in amorphous Al_2O_3 planar slab waveguides deposited by atomic layer deposition, ALD. For near UV applications, losses below 3 dB/cm have been demonstrated in high confinement single transverse mode ALD-grown and fully etched Al_2O_3 waveguides. These results demonstrate the wide wavelength range available to an Al_2O_3 based integrated photonics platform.

[0005] A major disadvantage of ALD-grown Al_2O_3 layers, when compared to layers grown using reactively sputtered Al_2O_3 , is the order of magnitude difference in deposition rate. Slab losses of sputtered Al_2O_3 reported to date are known as low as 0.1 dB/cm with fully etched waveguide losses below 0.2 dB/cm in the C-band. Therefore, a reduction of the losses in sputtered Al_2O_3 is required to compete with both ALD-grown Al_2O_3 , as well as the more established silicon nitride-based platform with demonstrated losses as low as 1 dB/m at 1550 nanometre for high confinement waveguides.

[0006] Known from the art is that a waveguide may be manufactured by providing a substrate, depositing an aluminium oxide waveguide core on the substrate, and arranging a cladding layer on the deposited aluminium oxide waveguide core. Here, it is noted that arranging the cladding layer comprises at least one processing step during which the deposited aluminium oxide waveguide core is subjected to a given maximum temperature.

[0007] It is known from the art that an optical waveguide with an aluminium oxide waveguide core made of amorphous aluminium oxide tends to show a high optical performance and/or low losses. At the same time, it is known that when deposited amorphous aluminium oxide waveguide cores are subjected to high temperatures, such as those occurring during annealing, optical performance markedly deteriorates.

[0008] The skilled person will be aware that optical losses in an optical waveguide are determined, not just by losses in the waveguide core, but also by losses in the cladding layer.

[0009] Optical losses in cladding layers may vary between different materials. For example, Tetraethyl orthosilicate, TEOS, is known to display relatively low optical losses.

However, for the manufacturing of such a TEOS cladding layer, a high temperature annealing step is typically required. When combining the TEOS cladding layer with an amorphous aluminium oxide waveguide core, a problem therefore arises in that the benefit of the optical quality of the TEOS cladding layer is mitigated by the losses in optical performance of the amorphous aluminium oxide waveguide core due to the high temperatures involved when using a TEOS cladding layer on the amorphous aluminium oxide waveguide core.

[0010] There may also be other types of cladding layers that, to be arranged on the deposited aluminium oxide waveguide core, require processing steps during which said aluminium oxide waveguide core is subjected to high temperatures, resulting in the aforementioned loss in optical performance of the aluminium oxide waveguide cores. Hereinafter, cladding layers that require a high-temperature step performed at 800°C . or higher during or after deposition are referred to as high-temperature cladding layers.

[0011] Accordingly, a problem exists when combining a known aluminium oxide waveguide core with a high-temperature cladding layer. This problem prevents further reductions in optical losses for aluminium oxide waveguide cores. Consequently, high-temperature cladding layers cannot be combined with known amorphous aluminium oxide waveguide cores, even though both components by themselves show high optical performance and/or low losses, and thus seem to be advantageous to use in combination.

[0012] This problem, for example, occurs in WEST, Gavin N., et al. Low-loss integrated photonics for the blue and ultraviolet regime. *Appl Photonics*, 2019, 4,2:026101. This paper discusses amorphous aluminium oxide, or alumina, films deposited using atomic layer deposition. The alumina films are grown either on bare silicon substrates or on thick ($3.2\text{ }\mu\text{m}$) silicon dioxide on silicon (also known as thermal oxide), at 300°C . growth temperature. Optical loss in the films was measured using a prism coupling method (Metricron) for wavelengths of 633 nanometres and 405 nanometres. Before annealing, optical loss was measured at less than 0.3 decibel per centimetre (dB/cm) or 30 decibel per metre (dB/m).

[0013] The authors indicate that high temperature annealing causes formation of dense polycrystalline γ -phase Al_2O_3 . High temperatures being above 800°C . The disclosed alumina films, after being annealed at 900°C . and 1100°C ., are polycrystalline and are said to exhibit optical losses of more than 20 dB/cm, or 2000 dB/m.

[0014] And while this problem is exemplified here for aluminium oxide waveguide cores, the applicant finds that while manufacturing aluminium nitride (AlN) waveguide cores, the same or at least very similar behaviour and problems occur.

[0015] An object of the present invention is to provide a method for manufacturing an optical waveguide comprising an aluminium nitride waveguide core and high-temperature cladding layer in which at least one of the abovementioned problems is at least partially solved.

[0016] According to the present invention, this object is achieved with a method as defined in claim 1, which is characterized in that depositing the aluminium nitride waveguide core comprises forming nano-crystallites in the aluminium nitride waveguide core, wherein a temperature the aluminium nitride waveguide core would be required to have for significantly increasing a size of the formed nano-

crystallites during arranging of the cladding layer exceeds the given maximum temperature, wherein the given maximum temperature is about 800° C. or higher.

[0017] In some embodiments, an increase in size is considered significant when it is an increase of about 100% or more. At the same time, any increase in size is detrimental, so in preferred embodiments, an increase is significant when between about 100% and about 50%.

[0018] The Applicant has found that by deliberately allowing nano-crystallites to form during deposition, an aluminium nitride waveguide core can be manufactured that has similar optical performance as one made of amorphous material, while being able to maintain said optical performance up to higher temperatures during subsequent processing steps.

[0019] The Applicant found that Rayleigh scattering plays an important role in deposited aluminium nitride layers. Firstly, Rayleigh scattering may occur as a result of differences in dielectric constant inside the aluminium nitride layer. Secondly, Rayleigh scattering may occur when crystallites are formed having a size that is at least comparable to the wavelength of the light in the layer. Other sources of optical scattering become relevant when crystallites have an even larger size.

[0020] When depositing at relatively low temperatures, no significant crystallization occurs, an amorphous layer will be achieved, and the optical losses of such layer will be relatively low. However, when such an amorphous aluminium nitride layer is subsequently subjected to a high temperature, e.g. 800° C. and up, crystallization will occur. In the amorphous material, crystallites will start to grow. Since these crystallites grow in amorphous material, they can grow freely, e.g. amorphous material surrounding a crystallite can be absorbed into this crystallite with relatively little energy, and in every direction. The resulting crystallites will be relatively large and have different dielectric properties from the remaining amorphous material. Optical scattering including Rayleigh scattering will therefor occur and optical performance will be degraded.

[0021] When depositing at moderately high temperatures, crystallization occurs and crystallites will be formed in an otherwise amorphous layer. In this case, local differences in the dielectric constant between the amorphous and nano-crystalline parts of the layer will exist as well. Rayleigh scattering associated with these differences will result in reduced optical performance. Furthermore, when this layer is subjected to high temperatures, for example during an annealing step, the existing nano-crystallites will grow by absorbing the amorphous material that surrounds them thus causing relatively large crystallites and increased optical scattering losses. When depositing at relatively high temperatures, crystallization will occur to such an extent that almost the entire deposited layer is filled with nano-crystallites. In this case, local differences in the dielectric constant will be largely absent as the amount of remaining amorphous aluminium nitride is small. Optical scattering in such layers is relatively low.

[0022] Without being bound by theory, the Applicant notes that the energy required for transforming the abovementioned aluminium nitride layer with a large amount of relatively small crystallites into a layer with a small amount of relatively large crystallites is too high to be reached during the processing steps required to arrange a cladding layer. That such a waveguide core is achieved can be

confirmed by subjecting it to the given maximum temperature. No significant changes in the optical performance will occur, as very little crystal growth will occur inside the layer.

[0023] By deliberately allowing nano-crystallites to form, no to little amorphous material is present. This means fewer local differences in dielectric properties, so less optical scattering. Also, there is less amorphous material for nano-crystallites to absorb and grow. Hence, in the aluminium nitride waveguide core deposited according to the invention, existing nano-crystallites cannot grow as easily and as freely, thereby significantly limiting their growth and avoiding the aforementioned optical scattering.

[0024] Without being bound by theory, the Applicant considers that the energy required for the growth in size of nano-crystallites that are surrounded predominantly by other nano-crystallites is larger than energy required for the growth in size of nano-crystallites that are surrounded predominantly by amorphous material. This is a further reason why, when the deposition of the aluminium nitride waveguide core is performed such that nano-crystallites are formed, formation of larger crystallites during subsequent processing steps can be prevented or at least limited. This allows for arranging a high-temperature cladding layer on the aluminium nitride waveguide core without significant degradation of the optical performance of the aluminium nitride waveguide core. As a result, waveguides having an aluminium nitride waveguide core can be manufactured that display improved optical performance and/or decreased losses.

[0025] The size of the nano-crystallites formed during deposition strongly impacts the amount of optical scattering. The Applicant found that by forming the nano-crystallites such that the given maximum temperature lies in a range between 800 and 1400° C., advantageous optical performance of the optical waveguide can be obtained. A maximum temperature that is lower than 800° C. would indicate that a relatively large amount of amorphous aluminium nitride would be available after deposition of the waveguide core that would transform in large crystallites when arranging the cladding layer. On the other hand, a maximum temperature that is higher than 1400° C. would risk deterioration of the aluminium nitride waveguide core, or other components included in the waveguide comprising said aluminium nitride waveguide core.

[0026] It is noted that the given maximum temperature more preferably lies in a range between 1000 and 1300° C. and is more preferably about 1150° C.

[0027] Arranging the cladding layer may comprise depositing the cladding layer on the aluminium nitride waveguide core and said at least one processing step may comprise annealing the combination of the substrate, deposited aluminium nitride waveguide core, and deposited cladding layer at the given maximum temperature.

[0028] The nano-crystallites in the aluminium nitride waveguide core after the deposition and before the annealing can be between 1 nanometre and 30 nanometre in size, and preferably between 1 nanometre and about 10 nanometre in size. Furthermore, the nano-crystallites in the aluminium nitride waveguide core may form at least 40% by weight of the aluminium nitride waveguide core after the deposition and before the annealing, preferably at least 70%, and more preferably at least 90%. Additionally or alternatively, the nano-crystallites in the aluminium nitride waveguide core after the annealing can be between 1 nanometre and 30

nanometre in size, and preferably between 1 nanometre and 10 nanometres. Furthermore, the nano-crystallites in the aluminium nitride waveguide core may form at least 50% by weight of the aluminium nitride waveguide core after the annealing, preferably at least 75%, and more preferably at least 99%.

[0029] The at least one processing step may comprise depositing the cladding layer on the aluminium nitride waveguide core at the given maximum temperature. In this embodiment, the high-temperature heating step is applied during the deposition of the cladding layer itself. The nano-crystallites in the aluminium nitride waveguide core after the depositing the cladding layer can be between 1 nanometre and 30 nanometre in size, and preferably between 1 nanometre and 10 nanometres. Furthermore, the nano-crystallites in the aluminium nitride waveguide core may form at least 50% by weight of the aluminium nitride waveguide core after the annealing, preferably at least 75%, and more preferably at least 99%.

[0030] The aluminium nitride waveguide core may be deposited using any one of reactive sputter deposition, atomic layer deposition, evaporation, or pulsed laser deposition. Other means of arranging the aluminium nitride waveguide core are not excluded.

[0031] The cladding layer may comprise high-temperature cladding layer. Such cladding layers are characterized in that these layers are arranged at relatively high temperatures and/or require heating steps after deposition at relatively high temperatures when compared to a temperature at which the aluminium nitride waveguide core is deposited. Examples of such cladding layers are a TEOS layer, a Silicon oxynitride layer, an aluminium oxide layer, or a polymer layer. Such layers have relatively low optical losses when compared to low-temperature cladding layers. The high-temperature cladding layer can be arranged using any one of plasma-enhanced vapor deposition, low-pressure chemical vapor deposition, evaporation, sputtering, or atomic layer deposition.

[0032] The substrate may comprise a silicon substrate, silicon nitride substrate, a silicon thermal oxide substrate, a quartz substrate, or a sapphire substrate.

[0033] The aluminium nitride waveguide core may be stoichiometric. Additionally or alternatively, the aluminium oxide waveguide core may comprise Al_xN_y , wherein $0.8 < x < 1.1$ and $0.9 < y < 1.2$, such as $x=0.9$, and $y=1.1$, and preferably $x=1$, and $y=1$.

[0034] The optical waveguide can be a slab waveguide or channel waveguide. The present application does not exclude other types of waveguides.

[0035] In further embodiments, depositing the aluminium nitride waveguide core on the substrate may further comprises one or more of the steps of reducing surface roughness of the aluminium nitride waveguide core, for example using chemical mechanical polishing, and defining a shape and/or size of the aluminium nitride waveguide core using at least one of lithography and etching.

[0036] The method may further comprise the step of, at a deposition rate for aluminium nitride, depositing aluminium nitride at varying substrate temperatures and/or varying substrate bias voltages on respective substrates. For each deposited aluminium nitride layer, its optical performance is measured. The method may further comprise selecting, as optimal settings, said deposition rate, substrate temperature and substrate bias voltage with which the aluminium nitride

layer was manufactured that had the best optical performance. These optimal settings can be used when depositing the aluminium nitride waveguide core for manufacturing an optical waveguide as described above.

[0037] According to a second aspect, the present invention provides an aluminium nitride optical waveguide that comprises a substrate, an aluminium nitride waveguide core arranged on the substrate, and a cladding layer arranged on the aluminium nitride waveguide core. According to the present invention, the aluminium nitride waveguide core comprises nano-crystallites that are between 1 nanometre and 30 nanometre in size, preferably between 1 nanometre and 10 nanometre in size. The nano-crystallites form at least 50% by weight of the aluminium nitride waveguide core, preferably at least 75%, and more preferably at least 99%, and the cladding layer comprises a high-temperature cladding layer. The high-temperature cladding layer may comprise at least one of a TEOS layer and a Silicon oxynitride layer. Additionally or alternatively, the optical waveguide is a slab waveguide, or a channel waveguide.

[0038] Next, the present invention will be described by referring to the appended drawings, wherein identical reference signs will be used to refer to the same or similar components, and wherein:

[0039] FIG. 1 shows a flowchart of a method for manufacturing an optical waveguide in accordance with the present invention;

[0040] FIG. 2 shows a cross-section of a slab waveguide in accordance with the present invention;

[0041] FIG. 3 shows a flowchart of a preferred embodiment of the method of FIG. 1;

[0042] FIG. 4 shows an example of a reactive co-sputtering system that may be configured to deposit an aluminium oxide waveguide core;

[0043] FIG. 5 illustrates TEM pictures of deposited aluminium oxide waveguide cores for various substrate temperatures;

[0044] FIGS. 6a and 6b illustrate AFM pictures of a deposited aluminium oxide waveguide cores for various substrate temperatures;

[0045] FIG. 7 shows a graph relating, on the X-axis, deposition temperatures for aluminium oxide waveguide cores, and, on the Y-axis, measured refractive index thereof, and a graph relating, for a number of deposition temperatures, on the X-axis, a wavelength of light propagated through aluminium oxide waveguide cores deposited at said temperatures, and, on the Y-axis, measured propagation losses therein.

[0046] FIG. 8a shows a graph relating, on the X-axis, the temperature at which a waveguide core is deposited, to, on the Y-axis, the size of crystallites formed in said aluminium oxide waveguide core;

[0047] FIG. 8b shows a graph relating, on the X-axis, the temperature at which a waveguide core is deposited, to, on the Y-axis, the percentage by weight of the waveguide core that is in a particular phase, specifically amorphous (A) or crystalline (C);

[0048] FIG. 8c shows a graph relating, on the X-axis, the temperature at which a waveguide core is deposited, to, on the Y-axis, optical losses achieved in said waveguide core;

[0049] FIGS. 9a and 9b both show graphs relating, on the X-axis, the distance over which light propagated through a waveguide core, to, on the Y-axis, intensity of said light.

[0050] FIG. 1 shows a flowchart including steps S1-S3 of one embodiment of a method for manufacturing an optical waveguide according to the present invention. FIG. 2 shows a schematic cross-section of an optical waveguide, as, for example, can be manufactured with the method according to the invention. FIG. 3 shows a flowchart of a preferred embodiment of said method.

[0051] In a step S1, a substrate 10 is provided. In a next step S2, an aluminium oxide waveguide core 11 is deposited on the substrate. The aluminium oxide waveguide core 11 may be deposited directly onto the substrate. Embodiments are also conceivable in which the aluminium oxide waveguide core 11 is deposited on the substrate, with other layers in between. An exemplary embodiment is one in which waveguide cores are stacked. A further exemplary embodiment is one in which the aluminium oxide waveguide core 11 is deposited on an aluminium oxide layer that, in turn, is deposited on the substrate. A further exemplary embodiment is one in which the substrate is a sapphire, preferably crystalline sapphire substrate and in which an SiO₂ layer is arranged on the substrate and in which the aluminium oxide waveguide core is deposited on the SiO₂ layer. Specifically, aluminium oxide waveguide core 11 is deposited such that nano-crystallites are formed therein. In a next step S3, a cladding layer 12 is arranged on aluminium oxide waveguide core 11. During at least one processing step required to arrange cladding layer 12, aluminium oxide waveguide core 11 is subjected to a given maximum temperature, which is about 800° C. or higher, and preferably between 800 degrees and 1400° C. The temperature the aluminium oxide waveguide core would be required to have for significantly increasing the size of the formed nano-crystallites during arranging the cladding layer exceeds this given maximum temperature. Alternatively said, the lowest temperature at which the formed nano-crystallites will significantly increase in size, exceeds the given maximum temperature. Aluminium oxide waveguide core 11 can therefore have said maximum temperature without said significant increase in size occurring.

[0052] Increasing the maximum temperature beyond 1400° C. risks deterioration of aluminium oxide waveguide core 11. The skilled person will appreciate that around 1400 degrees Celsius the glass transition temperature of aluminium oxide starts, so subjecting the waveguide core to this temperature may cause reflow of the structure. Moreover, such a maximum temperature may also deteriorate the performance of the other components of the waveguide, depending on the materials used. Substrate 10 may be made of silicon, which may start to melt around 1400° C. Substrate 10 may also comprise silicon dioxide, in which, when exposed to such temperatures, local density fluctuations may form and these cause scattering, thus degrading performance.

[0053] The mentioned increase in size of said crystallites may be considered significant when about 100% or more, preferably when between about 100% and about 50%.

[0054] Now referring to FIG. 3, step S3 of arranging the cladding layer may comprise a step S31 of depositing a cladding layer 12 on aluminium oxide waveguide core 11. Step 31 may be the processing step during which aluminium oxide waveguide core 11 is subjected to the given maximum temperature. This may for example apply in embodiments where a polymer-based cladding layer is deposited.

[0055] Alternatively, a high-temperature step is not part of the deposition of the cladding layer but part of subsequent heating step. In FIG. 3, an example of such subsequent step is provided as step S32 in which the combination of substrate 10, deposited aluminium oxide waveguide core 11, and arranged cladding 12 layer is annealed thereby subjecting aluminium oxide waveguide core 11 to the given maximum temperature. This applies for example in embodiments where a TEOS cladding layer is deposited.

[0056] It should be noted that various other steps can be performed in between the deposition of aluminium oxide waveguide core 11 and the arrangement of cladding layer 12. For example, the deposited aluminium oxide layer can be subjected to chemical mechanical polishing to reduce surface roughness and to lithography and etching steps to define a channel waveguide or other type of waveguide.

[0057] For any embodiment (e.g. methods and/or devices) relating to aluminium oxide waveguide core 11, an alternative embodiment may be considered in which the waveguide core is instead made of aluminium nitride, AlN. The embodiments explained in relation to FIGS. 1-3, for example, can also be applied for manufacturing a waveguide comprising an aluminium nitride waveguide core. The skilled person will be aware of how the following specification of a manufacturing process for an aluminium oxide waveguide core, can and/or should be adapted for manufacturing an aluminium nitride waveguide core.

[0058] The Al₂O₃ thin films used in embodiments of the present invention can be deposited by reactive sputtering onto a silicon wafer having an 8 micrometres oxide buffer layer. The advantage of reactive sputtering exploited here is a result of the energy available per adatom on the substrate. The adatoms landing on the substrate have a high mobility, resulting in high-density layer morphologies available at relatively low substrate temperatures and high deposition rates. This allows for the deposition of high-density amorphous Al₂O₃ layers at CMOS compatible wafer temperatures, with slab waveguide propagation losses below 0.1 dB/cm at 1550 nm.

[0059] While several low-loss propagation slab waveguide results for Al₂O₃ optical waveguides have been achieved, no link with layer morphology has been demonstrated in the prior art. The Applicant realized that, given the complexity of reproducing reactive sputter deposition processes, an understanding of the morphology and corresponding propagation losses can facilitate improved reproducibility and layer quality.

[0060] The morphology of an Al₂O₃ layer deposited by reactive sputtering is primarily determined by the available energy per adatom, EPA, and the material properties of the deposited layer. The material properties determining the morphology are the activation energy and diffusion constants that determine the diffusion length of the adatom at a given kinetic energy available per adatom, and the critical nucleation dimension that determines a critical diffusion length required for stable nucleation. While the material properties are a given for an Al₂O₃ layer, the energy per adatom is a ratio of the deposition rate and the total energy contributions during deposition.

[0061] For reactive sputtering, the total energy flux is a linear combination of different contributions. The sputtering process contributions to the energy flux towards the substrate can be classified in at least four groups.

[0062] The contributions of the atoms and molecules adhering to the substrate required for Al_2O_3 layer formation should be considered first. Adatoms that have been accelerated from the target have contributed their kinetic energy as they were adsorbed on the substrate. In addition, when an oxygen molecule is adsorbed, its kinetic energy is also contributed. Even when atoms or molecules are not adsorbed, part of their kinetic energies can still be transferred when colliding with the substrate. Especially when the gas molecules get energized by collisions with higher velocity ions, this contribution can become significant. Another form of kinetic energy is related to the temperature of the substrate. Furthermore, besides the kinetic energy contributions to the layer formation, the potential energy released by the exothermic chemical reaction forming Al_2O_3 is a significant contribution.

[0063] The three other groups of energy contributions are radiation from the plasma, electrons incident on the substrate, and ions accelerated towards the substrate. Note, that a substrate bias can be applied. Increasing and/or decreasing said bias can be done to increase or decrease the electron and ion bombardment on the substrate.

[0064] All the energy contributions add to the available energy per adatom and thus influence the layer morphology and consequent propagation losses.

[0065] In an embodiment, the Al_2O_3 layer can be deposited using an AJA ATC 1500 RF reactive co-sputtering system **100** on 10 centimetre silicon wafers having an 8 micrometre thick thermal oxide buffer layer.

[0066] System **100**, which is schematically illustrated in FIG. 4, comprises a target **101** comprising aluminium with a 99.9995% purity, which is arranged above cathode **102**. Opposite to target **101**, substrate **10** is arranged on anode **103**, which is electrically connected to chamber **103A**. RF power is applied between anode **103** and cathode **102**. This causes the generation of a plasma **104** in which supplied Ar atoms **105** are ionized into Ar ions **106** and electrons **107**. Ar ions **106** are accelerated towards target **101** under the influence of a self-generated DC bias. At target **101**, they will collide with Al atoms thereby generating a stream of Al atoms **108** towards substrate **10**. At substrate **10**, Al atoms **108** that have deposited onto substrate **10** will react with oxygen molecules **109** to form aluminium oxide.

[0067] The main deposition chamber is evacuated through inlet **110** to a base pressure of 0.1 microTorr to prevent incorporation of hydroxide ions in the Al_2O_3 layer, which induce absorption losses around 750 nm, 970 nm, and 1400 nm.

[0068] For the magnetron discharge to be sustained, an equilibrium between the secondary electrons emission from target **101** under ion bombardment and the rate of electrons **107** escaping plasma **104** needs to be maintained. While an RF power source does not directly apply a DC potential difference between cathode **102** and anode **103**, electrons **107** in plasma **104** absorb the RF energy much more efficiently than heavier argon ions **106**. The high electron mobility causes electrons **107** to be collected on the electrodes. A self-generated DC bias voltage is then a consequence of the asymmetry between target **101** and chamber **103A** of sputtering system **100**. A magnetic field is applied using permanent magnets below the target to increase the electron density in plasma **104**, thereby increasing the argon ionization rate and reducing the discharge voltage required.

In addition, the magnetic field greatly increases the sputter yield by the increase of ionization and therefore bombardment rate of the target.

[0069] Sputtering system **100** further comprises heating means, such as infrared heaters **111**, for heating substrate **10** either directly or via anode **103** on which substrate **10** is arranged.

[0070] Exemplary process conditions for depositing the aluminium oxide waveguide core are listed in the table below.

Parameter	Value	Unit
Base pressure	<0.3	μTorr
Argon flow	30	Sccm
Oxygen flow	3.0 ± 0.2	Sccm
Oxygen partial pressure	0.2 ± 0.1	mTorr
Pressure	3.5 ± 0.1	mTorr
Pump flow	50 ± 2	l/s
Substrate set temperature	420 460 500 540 580 620	$^{\circ}\text{C}$.
	660 700	
Target-substrate distance	15.2	cm
Power	200	W

[0071] Given the dependence of the layer morphology on available energy per adatom, it is possible to vary the substrate temperature for the purpose of varying the available energy per adatom as it is mostly independent of the other parameters in the process. This allows the investigation of layer morphology as it changes with substrate temperature and the corresponding optical propagation losses in the layers.

[0072] The substrate temperatures are set temperatures measured on the substrate holder and are therefore not the exact temperature of the substrate. A calibration of substrate temperature as a function of the set temperature can be provided.

[0073] FIG. 5 illustrates TEM pictures of a deposited aluminium oxide waveguide layer for various substrate temperatures. The morphology of the layer at the lowest chosen temperature of 420°C . is amorphous. As the substrate temperature is increased to 460°C . nano-crystallites start to form, of which the density is significantly increased for 500°C . and 540°C .

[0074] As the temperature is increased from 500 to 580°C ., the surface roughness increases with ever larger waviness. While the waviness of the surface is still present for the layer deposited at 580°C ., a clear transition from a mostly amorphous layer with nano-crystallites to a mostly polycrystalline morphology has occurred. The waviness disappears from a temperature of 620°C . as an increasingly polycrystalline morphology is observed with a slight columnar growth profile. As the temperature is further increased to 700°C ., no significant difference in the morphology is observed.

[0075] FIGS. 6a and 6b illustrate AFM measurements of a deposited aluminium oxide waveguide layer for various substrate temperatures. The AFM measurements show the waviness appearing at 500°C . (FIG. 6a, bottom left), increasing in amplitude at 540 degrees Celsius (FIG. 6a, bottom right), and reducing and disappearing at 620°C . (FIG. 6b, top right). In addition to the waviness, the layers grown at a temperature from 500°C . (FIG. 6a, bottom left) to 580°C . (FIG. 6b top left) exhibit a decreased refractive index and thickness uniformity.

[0076] FIG. 7, bottom, specifically shows the refractive index of aluminium oxide layers measured by ellipsometry at 1550 nm, for varying deposition temperatures.

[0077] The optical propagation losses for each aluminium oxide layer have been investigated using a Metricon 2010/M41, with a fibre loss module. These losses are illustrated in figure, top. Clearly, losses decrease with increasing deposition temperature down to 1.57 dB/cm at 377 nm and 0.84 dB/cm at 403 nm for the layer grown at 700° C. substrate temperature.

[0078] In alternative embodiments, when depositing an aluminium nitride waveguide core, the exact deposition temperatures may be different from those mentioned in relation to FIGS. 5 and 6, however the applicant does find that the same behaviour occurs. The same changes in morphology, and/or the same trends in refractive index and thickness uniformity can be observed in aluminium nitride waveguide cores, when deposited at varying temperatures. The teachings derived from FIGS. 4-7, while explaining based on the aluminium oxide waveguide core embodiment, can also be applied to the aluminium nitride waveguide core embodiment.

[0079] FIGS. 8a-c each show a graph that relates the deposition temperature at which an aluminium oxide waveguide core 11 is deposited, to various properties of said core 11. On the X-axis of each of these graphs the deposition temperature is shown. The skilled person will appreciate that the exact temperature values at which a particular waveguide core is deposited is strongly machine dependent, e.g. the machine given value for the 'deposition temperature' can deviate from an actual, practically very hard to know, temperature of the waveguide core. However, given a particular machine and a deposition rate that can be achieved, a temperature sweep can be performed to determine exact temperature values at which the behaviour as elucidated in the graphs shown in FIGS. 8a-c occurs.

[0080] In the art, there is a clear preference for making dense amorphous Al_2O_3 waveguide cores as these cores have displayed low optical losses. In FIGS. 8a-8c, the deposition temperature at which such waveguides are achieved is referred to as temperature P1. If, starting from P1, the deposition temperature is decreased, the deposited layer will become less dense and voids will be present in the amorphous aluminium oxide. Said voids may act as scattering objects and may cause losses. If, starting from P1, the deposition temperature is increased, nano-crystallites form in the amorphous material. Said nano-crystallites may act as scattering objects and may cause losses.

[0081] As shown in FIGS. 8a-8c, P1 is a local minimum at which a balance is struck between reducing the amount voids and preventing the formation of nano-crystallites. At P1, relatively low losses can be achieved. Known aluminium oxide optical waveguides are based on aluminium oxide layers deposited at temperatures corresponding to P1. However, such layers have the aforementioned disadvantage of being vulnerable to high temperature processing steps that follow the deposition of the aluminium oxide layer.

[0082] The Applicant realised that in aluminium oxide optical waveguides deposited at deposition temperatures in a range around P1, the largest losses are caused by local differences in dielectric properties. Both voids and crystallites have different dielectric properties than amorphous

aluminium oxide. More voids and/or more crystallites means such local differences occur more often and optical losses will increase.

[0083] The Applicant further realised that such local differences will most often occur, and therefore that the losses caused thereby may be at their highest, when the aluminium oxide layer comprises comparable amounts of amorphous material and nano crystallites. This may for example be seen in FIG. 8b, in which line A describes how much aluminium oxide in the layer, expressed in weight percentage, is in the amorphous phase, and line C describes how much aluminium oxide, expressed in weight percentage, is in the crystalline phase. The skilled person will appreciate that the amount of amorphous material and/or the number of nano-crystallites may also be expressed using other units and/or metrics.

[0084] The Applicant further realised that when the deposition temperature is increased, the nano-crystallites will at some temperature overtake the amorphous material as the dominant material in the deposited layer. As the number of discontinuities between amorphous and crystalline aluminium oxide decreases due to the decreasing amount of amorphous aluminium oxide, a reduction in optical losses can be observed beyond a temperature P2. In FIG. 8b, temperature P2 is chosen as the temperature at which the content of amorphous and crystalline aluminium oxide content is identical for illustrative purposes only.

[0085] The reduction in optical losses continues up to the deposition temperature at which substantially all aluminium oxide in the waveguide core takes the form of nano-crystallites and little to none of it is in the amorphous phase. This point may be referred to as P3. FIG. 8c further indicates a rectangle R that illustrates the temperature range corresponding to the temperature variation in FIG. 7, top.

[0086] Equivocally, for deposition temperatures between P2 and P3, it is the sporadic presence of volumes of amorphous material in between the otherwise nano-crystalline aluminium oxide that can be considered the cause of scattering. Hence, when further increasing the deposition temperature, less and/or smaller volumes of amorphous material are formed, local differences occur less often, and losses decrease.

[0087] The Applicant furthermore realised that, while in the range P1 and P3, more and more aluminium oxide takes on the shape of nano-crystallites, that a size of the individual nano-crystallites does not increase significantly. This is conceptually reflected in FIG. 8a. Only when deposition temperatures above P3 are used, there may be a significant increase in the size of the individual crystallites while the percentage of aluminium oxide included in crystallites in itself stays the same. It should be noted that no size is indicated for deposition temperatures below P1 as almost no nano-crystallites are present. Increasing the temperature beyond P3 will cause nano-crystallites to combine into large crystallites. These relatively large crystallites cause an increase in optical scattering, thereby increases the optical losses in the layer. As such, at temperature P3 a local minimum can be observed in the losses that is comparable to the minimum at temperature P1. However, unlike deposited aluminium oxide layers deposited at temperature P1, aluminium oxide layers deposited at temperature P3 are much less susceptible to subsequent heating steps, such as an annealing step for processing a deposited cladding layer.

[0088] The morphology achieved at deposition temperatures **P3** may also be described as polycrystalline aluminium oxide, saturated with nano-crystallites. Here, nano-crystallites refer to crystallites with a size that is relatively small compared to the waveguide of the light that is to travel through the optical waveguide. This morphology is advantageous because:

[0089] a) nano-crystallites are so small that they do not cause significant Rayleigh scattering due to their size;

[0090] b) saturation of the core with nano-crystallites ensures little to no fluctuations in dielectric properties throughout the core thereby also limiting Rayleigh scattering;

[0091] c) saturation of the core with nano-crystallites also means that existing nano-crystallites have little to no amorphous material in their surroundings to absorb and grow with;

[0092] d) growth occurring due to nano-crystallites mutually aligning and forming a single larger crystallite only occurs at much higher temperatures.

[0093] Graphs in FIGS. **8a-c** conceptually show when depositing an aluminium oxide layer at various substrate temperatures. While points **P1**, **P2**, and **P3** are indicated in each graph, the behaviour related therein does not have to occur exactly at the same temperatures for each property described in FIGS. **8a-c**. Also, while FIG. **8c** suggests the local minima at **P1** and **P3** allow for achieving the same low losses, implementations of the method according to the invention may, for a plethora of reasons, result in local minima **P1** and **P3** achieving different levels of losses.

[0094] The applicant finds that subjecting aluminium oxide waveguide core **11**, in which nano-crystallites are formed, to temperatures of 800 degrees or higher may even be advantageous and/or decrease losses. This is for example shown in FIGS. **9a** and **9b**.

[0095] When aluminium oxide waveguide core **11** is exposed to such a temperature, while the growth of the nano-crystallites is very limited, said growth will still consume all or at least most of the amorphous aluminium oxide that may have formed during deposition of aluminium oxide waveguide core **11**. After being subjected to said temperature, if less amorphous aluminium oxide is present, local differences in dielectric properties occur less often and thus scattering decreases.

[0096] For FIG. **9a**, the aluminium oxide waveguide core in question is one deposited such that nano-crystallites are formed in the aluminium oxide. No further processing step was performed during which the deposited aluminium oxide waveguide core was subjected to a temperature of 800° C. or more. The losses achieved are 1+/-0.5 dB/cm.

[0097] For FIG. **9b**, the aluminium oxide waveguide core of FIG. **9a** was subjected to about 1150° C. for about four hours, in a nitrogen environment. The losses achieved are 0.7+/-0.2 dB/cm.

[0098] The intensity of light, as given on the Y-axis, is an estimate derived from scattering light measured over the propagation length of the waveguide core. Not willing to be bound by theory, the skilled person will appreciate that due to limitations of this estimate it may seem that the intensity of the light increases, while it is still safe to say that losses do occur. These losses are estimated by fitting the measurement data to a log-linear model using a maximum likelihood estimator sample consensus, MLESAC, algorithm. The given error margins are determined by fitting different

sections of the total propagation. The skilled person will appreciate that other approaches to estimate the intensity of light inside the waveguide core, and derive average losses from that measurement data may also be used.

[0099] In alternative embodiments, when depositing an aluminium nitride waveguide core, the temperature values for **P1**, **P2**, or **P3** may be different from those found for aluminium oxide waveguide cores, however the applicant does find that the same behaviour occurs. Similar changes in the phase that the aluminium nitride has (e.g. amorph or (nano-) crystalline), similar increase in size of the (nano-) crystallites, and a conceptually comparable loss profile can be identified in aluminium nitride waveguide cores, when deposited at varying temperatures. The teachings explained in FIGS. **8a-c**, and **9a-b**, while explain based on the aluminium oxide waveguide core embodiment, can also be applied to the aluminium nitride embodiment.

[0100] In the above, the present invention has been explained using detailed embodiments thereof. However, it should be apparent to the skilled person that various modifications are possible to these embodiments without deviating from the scope of the present invention, which is defined by the appended claims and their equivalents.

1. A method for manufacturing an optical waveguide, the method comprising:

providing a substrate;

depositing an aluminium nitride waveguide core on the substrate; and

arranging a cladding layer on the deposited aluminium nitride waveguide core, said arranging comprising at least one processing step during which the deposited aluminium nitride waveguide core is subjected to a given maximum temperature,

wherein depositing the aluminium nitride waveguide core comprises forming nano-crystallites in the aluminium nitride waveguide core,

wherein a temperature the aluminium nitride waveguide core would be required to have for significantly increasing a size of the formed nano-crystallites during arranging of the cladding layer exceeds the given maximum temperature, wherein the given maximum temperature is about 800° C. or higher, and

wherein the aluminium nitride waveguide core comprises nano-crystallites that are between 1 nanometre and 30 nanometre in size wherein the nano-crystallites form at least 50% by weight of the aluminium nitride waveguide core and wherein the cladding layer comprises a high-temperature cladding layer.

2. The method according to claim 1, wherein an increase in size of the formed nano-crystallites is significant when about 100% or when between about 100% and about 50%.

3. The method of claim 1, wherein the given maximum temperature lies in a range between 800 and 1400° C.

4. The method according to claim 1, wherein arranging the cladding layer comprises depositing the cladding layer on the aluminium nitride waveguide core, and wherein said at least one processing step comprises annealing the combination of the substrate, deposited aluminium nitride waveguide core, and deposited cladding layer at said given maximum temperature,

wherein the nano-crystallites in the aluminium nitride waveguide core after said deposition of the cladding layer and before said annealing are between 1 nanometre and 30 nanometre in size, and

wherein the nano-crystallites in the aluminium nitride waveguide core form at least 40% by weight of the aluminium nitride waveguide core after said deposition and before said annealing.

5. (canceled)

6. (canceled)

7. The method according to any of the claim 4, wherein the nano-crystallites in the aluminium nitride waveguide core after said annealing are between 1 nanometre and 30 nanometre in size,

wherein the nano-crystallites in the aluminium nitride waveguide core form at least 50% by weight of the aluminium nitride waveguide core after said annealing.

8. (canceled)

9. The method according to claim 1, wherein said at least one processing step comprises depositing the cladding layer on the aluminium nitride waveguide core at said given maximum temperature,

wherein the nano-crystallites in the aluminium nitride waveguide core after depositing the cladding layer are between 1 nanometre and 30 nanometre in size, and

wherein the nano-crystallites in the aluminium nitride waveguide core form at least 50% by weight of the aluminium nitride waveguide core after depositing the cladding layer.

10. (canceled)

11. (canceled)

12. The method according to claim 1, wherein the aluminium nitride waveguide core is grown using any one of reactive sputter deposition, atomic layer deposition, evaporation, or pulsed laser deposition, and/or

wherein the cladding layer comprises a TEOS layer, a Silicon oxynitride layer, an aluminium oxide layer, or a polymer layer,

wherein the cladding layer is arranged using any one of plasma enhanced vapor deposition, low pressure chemical vapor deposition, evaporation, sputtering, or atomic layer deposition, and

wherein the substrate comprises a silicon substrate, silicon nitride substrate, a silicon thermal oxide substrate, a quartz substrate, or a sapphire substrate.

13-15. (canceled)

16. The method according to claim 1, wherein the aluminium nitride waveguide core is stoichiometric, and/or wherein the aluminium nitride waveguide core comprises Al_xN_y , wherein $0.8 < x < 1.1$ and $0.9 < y < 1.2$.

17. The method according to claim 1, wherein the waveguide is a slab waveguide or a channel waveguide.

18. The method according to claim 1, further including, between depositing the aluminium nitride waveguide core and arranging the cladding layer, reducing surface roughness of the aluminium nitride waveguide core, for example using chemical mechanical polishing.

19. The method according to claim 1, further comprising defining a shape and/or size of the aluminium nitride waveguide core using, for example, at least one of lithography and etching, before arranging the cladding layer.

20. The method according to claim 1, comprising:

at a deposition rate for aluminium nitride, depositing aluminium nitride layers at varying substrate temperatures and/or at varying substrate bias voltages on respective substrates;

for each deposited aluminium nitride layer, measuring its optical performance;

selecting, as optimal settings, said deposition rate and the substrate temperature and substrate bias voltage with which the aluminium nitride layer was manufactured that had the best optical performance; and

using the optimal settings when depositing the aluminium nitride waveguide core for manufacturing an optical waveguide according to any of the preceding claims.

21. An aluminium nitride optical waveguide, comprising: a substrate;

an aluminium nitride waveguide core arranged on the substrate; and

a cladding layer arranged on the aluminium nitride waveguide core;

wherein the aluminium nitride waveguide core comprises nano-crystallites that are between 1 nanometre and 30 nanometre in size, wherein the nano-crystallites form at least 50% by weight of the aluminium nitride waveguide core, and wherein the cladding layer comprises a high-temperature cladding layer.

22. The aluminium nitride optical waveguide according to claim 21, wherein the high-temperature cladding layer comprises at least one a TEOS layer or a Silicon oxynitride layer, and/or

wherein the waveguide is a slab waveguide or a channel waveguide.

23. (canceled)

24. The method according to claim 1, wherein the nano-crystallites are between 1 nanometre and 10 nanometre in size.

25. The method according to claim 1, wherein the nano-crystallites form at least 75% by weight of the aluminium nitride waveguide core.

26. The method according to claim 1, wherein the nano-crystallites form at least 99% by weight of the aluminium nitride waveguide core.

27. The method according to claim 6, wherein the given maximum temperature lies in a range between 1000 and 1200° C.

28. The method according to claim 6, wherein the given maximum temperature is about 1150° C.

29. The aluminium nitride optical waveguide according to claim 21, wherein the aluminium nitride optical waveguide is manufactured by a method, the method comprising:

providing a substrate;

depositing an aluminium nitride waveguide core on the substrate; and

arranging a cladding layer on the deposited aluminium nitride waveguide core, said arranging comprising at least one processing step during which the deposited aluminium nitride waveguide core is subjected to a given maximum temperature,

wherein depositing the aluminium nitride waveguide core comprises forming nano-crystallites in the aluminium nitride waveguide core,

wherein a temperature the aluminium nitride waveguide core would be required to have for significantly increasing a size of the formed nano-crystallites during arranging of the cladding layer exceeds the given maximum temperature, wherein the given maximum temperature is about 800° C. or higher, and

wherein the aluminium nitride waveguide core comprises nano-crystallites that are between 1 nanometre and 30 nanometre in size, wherein the nano-crystallites form at least 50% by weight of the aluminium nitride waveguide core, and wherein the cladding layer comprises a high-temperature cladding layer.